Writing Inequalities from Verbal Constraints

Turning a limit in words into an inequality: at most and at least, more than and fewer than, and limits with a starting amount

Directions:

- Four phrases carry the whole sheet:

  at most / no more than $\to\leq$ **at least** / minimum $\to\geq$ **fewer than** / under $\to<$ **more than** / over $\to>$

- **The test-value model.** Before you solve, try numbers either side of the limit and mark which ones the situation allows. The boundary is the number where the answer changes, and whether the boundary itself is allowed is exactly what tells $\leq$ apart from $<$.

- For each problem: name the unknown with the letter given, write an inequality that says what the sentence says, then write the solution on the answer line (for example $x \leq 6$).

1. A lift may carry no more than 450 kilograms. Each box loaded into it weighs 25 kilograms. Let b be the number of boxes.

number of boxes b	17	18	19
mass of the boxes, $25b$ kg			
allowed by the 450 kg limit?			

Write the inequality for b and give its solution.

Answer: ________________

2. A film may be watched only by people over 12 years old. Let y be the age in years of a person who may watch it. Write the inequality for y and give its solution.

Answer: ____________________

3. A test is passed with at least 36 points. Each correct question is worth 4 points. Write an inequality for the number q of correct questions needed to pass, and give its solution.

Answer: ____________________

4. A taxi charges 4 dollars to start the trip plus 3 dollars for each kilometre. A passenger has at most 25 dollars to spend. Let k be the number of kilometres.

(a) Write the inequality for k in the work space, and copy it on this line: __________

(b) the solution: __________

5. A parcel must weigh less than 20 kilograms. The empty box and its packing already weigh 6 kilograms. Write an inequality for the mass m in kilograms that may still be added, and give its solution.

Answer: ___________________

6. To earn a badge, a scout must walk at least 60 kilometres in total. She has already walked 23 kilometres. Let d be the further distance in kilometres.

further distance d km	36	37	38
total walked, $23 + d$ km			
badge earned?			

Write the inequality for d and give its solution.

Answer: ___________________

7. A shelf can hold at most 45 kilograms. A pair of bookends already on it weighs 3 kilograms, and each book weighs $\frac{3}{2}$ kilograms. Write an inequality for the number n of books the shelf may hold, and give its solution.

Answer: ________________

8. Fewer than 24 players show up for a game, and they split into 6 equal teams. Write an inequality for the number p of players on one team, and give its solution.

Answer: ________________

9. A pool must hold fewer than 900 litres. It already holds 150 litres, and a tap adds 30 litres each minute. Let t be the number of minutes the tap runs.

(a) Write the inequality for t in the work space, and copy it on this line: _________

(b) the solution: _________

10. A bus tour costs 30 dollars for the driver plus 12 dollars for each person, and a group has at most 210 dollars. Let n be the number of people.

(a) Write the inequality for n in the work space, and copy it on this line: _________

(b) the solution: _________

(c) Explain why not every number in that solution set is a possible group size. Write your answer in the work space.